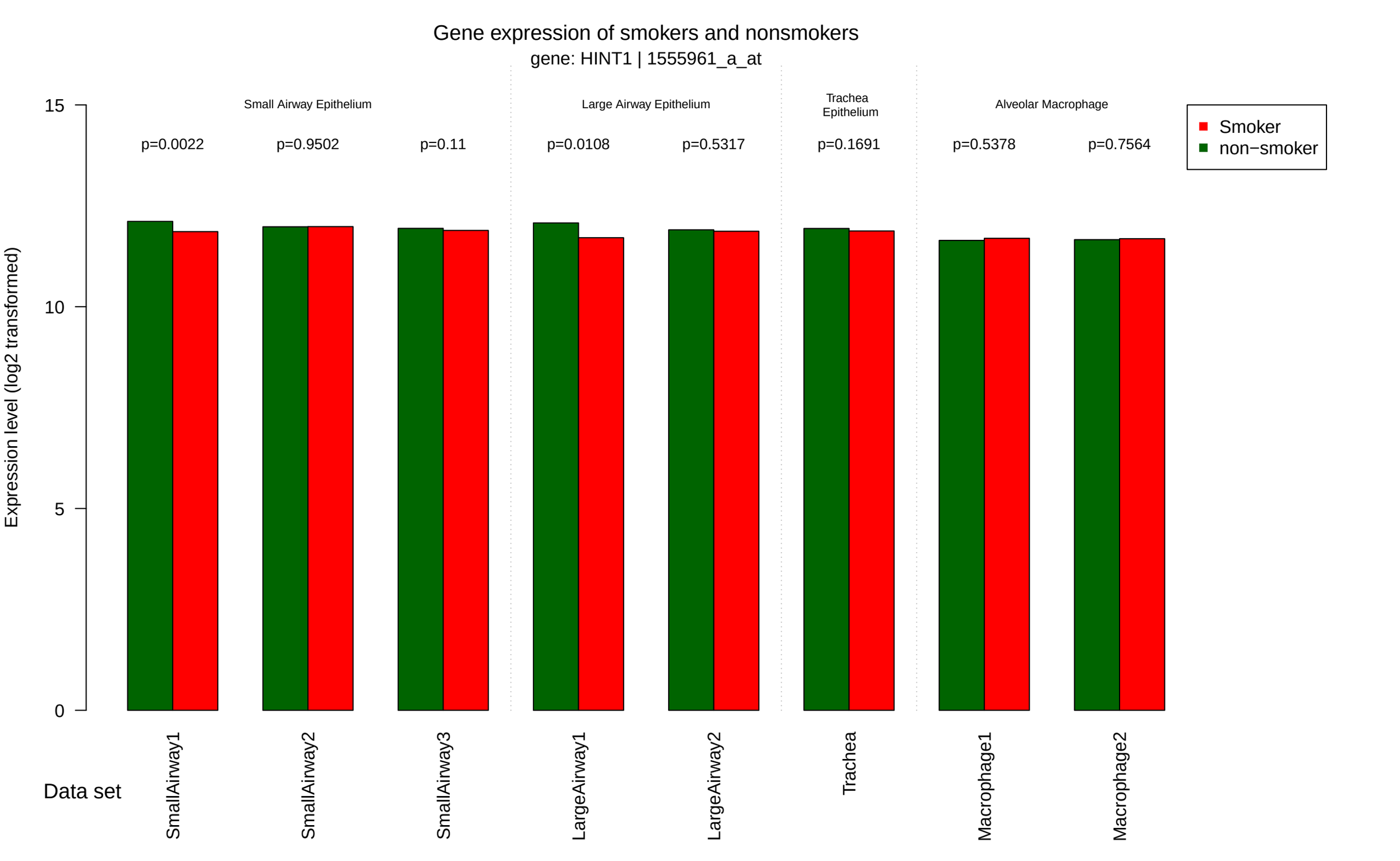
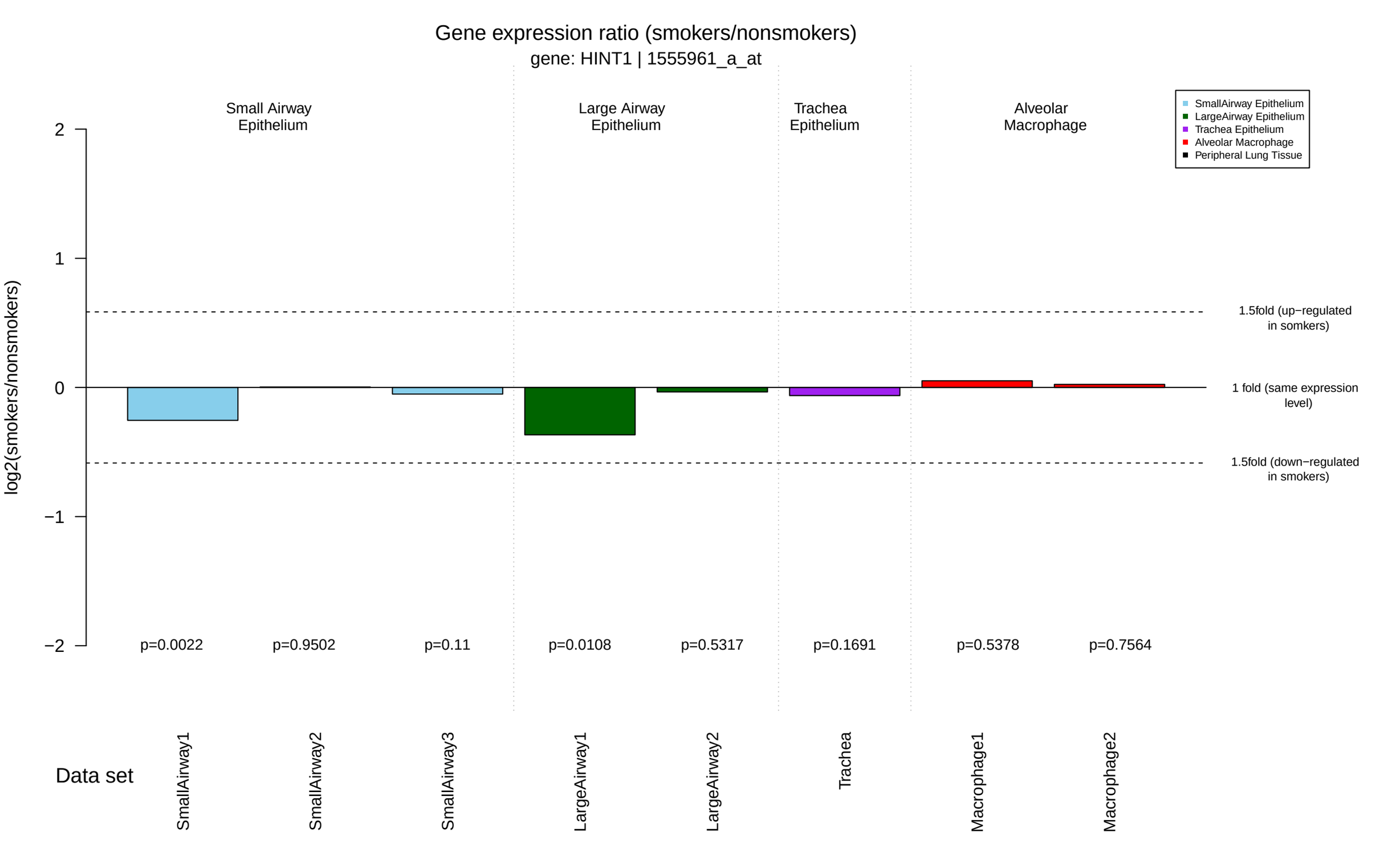
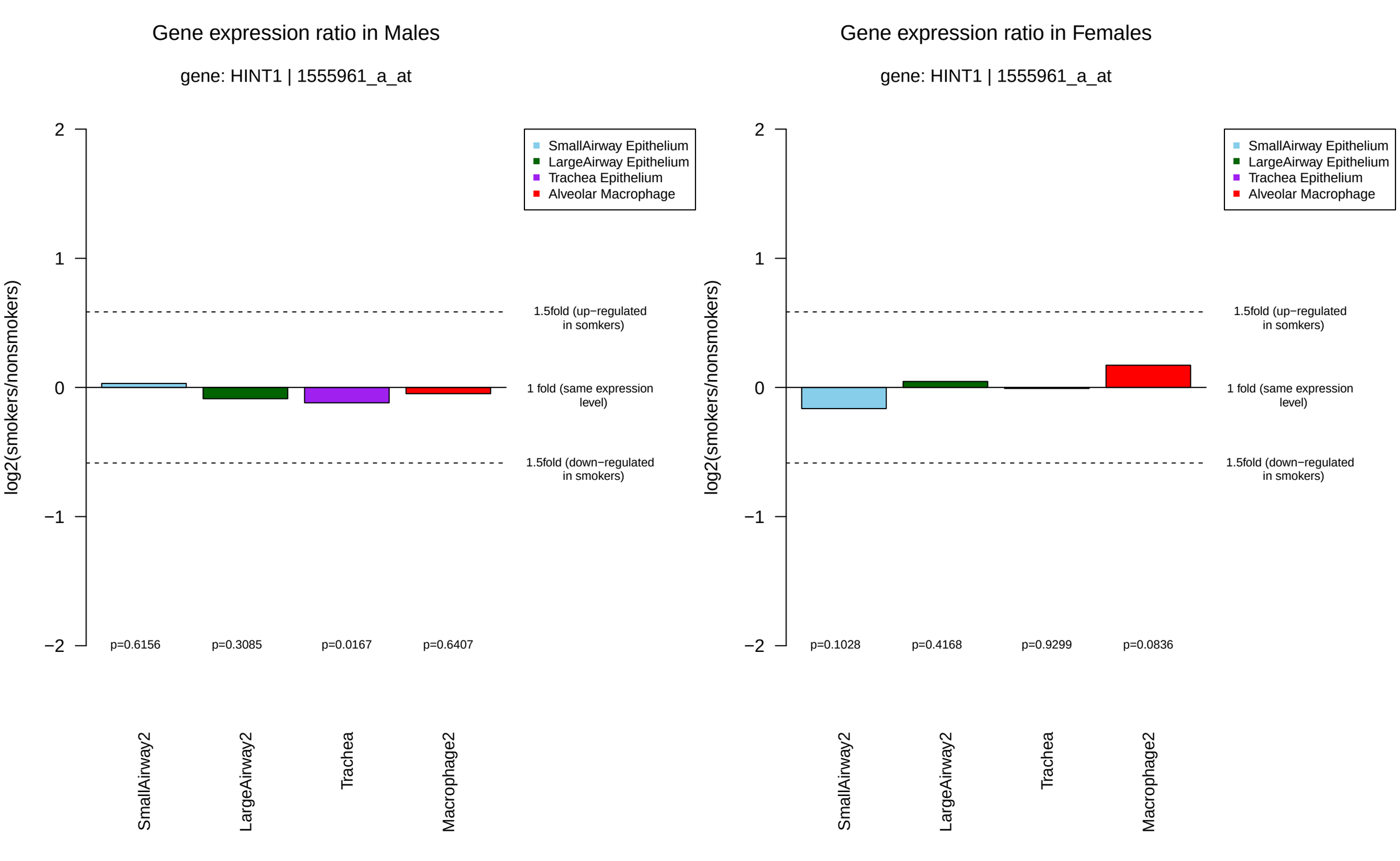
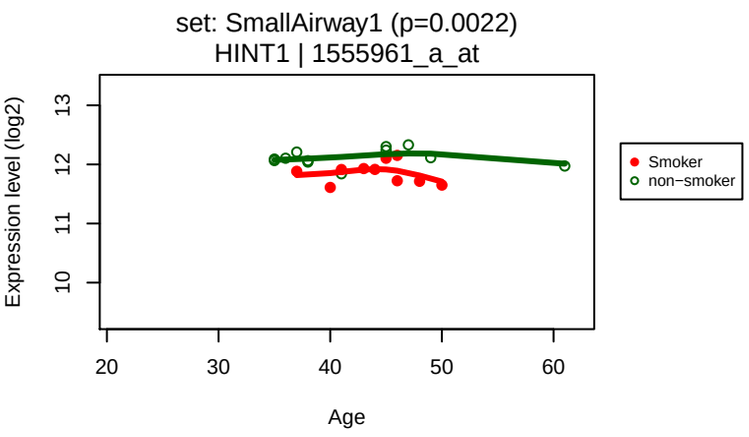
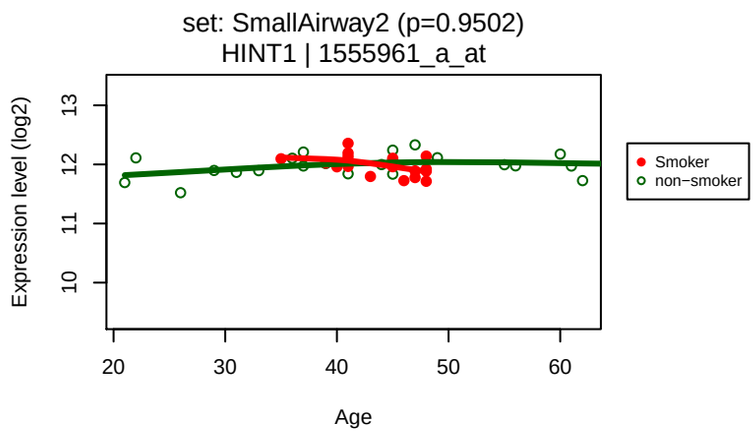
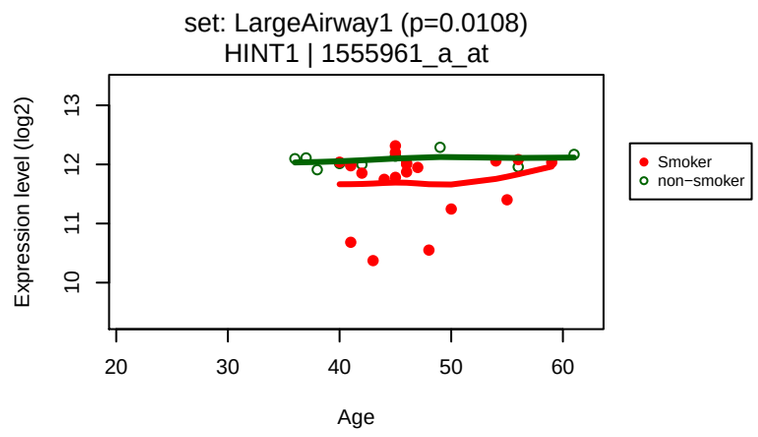

Gene expression of smokers and nonsmokers
gene: HINT1 | 1555961_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0022
p=0.9502
p=0.11
p=0.0108
p=0.5317
p=0.1691
p=0.5378
p=0.7564
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

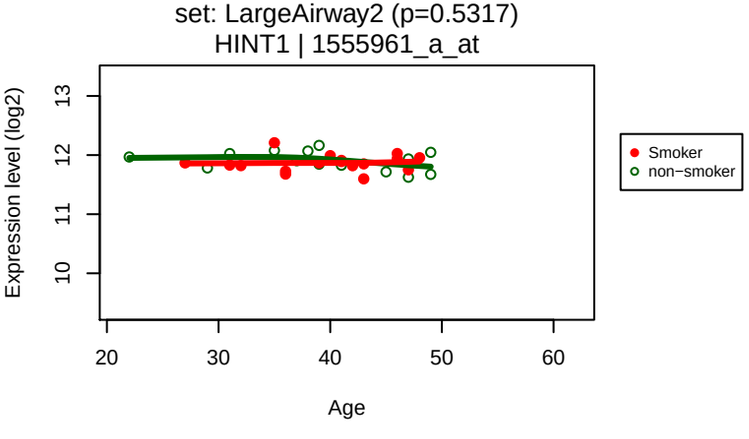

Gene expression ratio (smokers/nonsmokers)
gene: HINT1 | 1555961_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0022
p=0.9502
p=0.11
p=0.0108
p=0.5317
p=0.1691
p=0.5378
p=0.7564
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

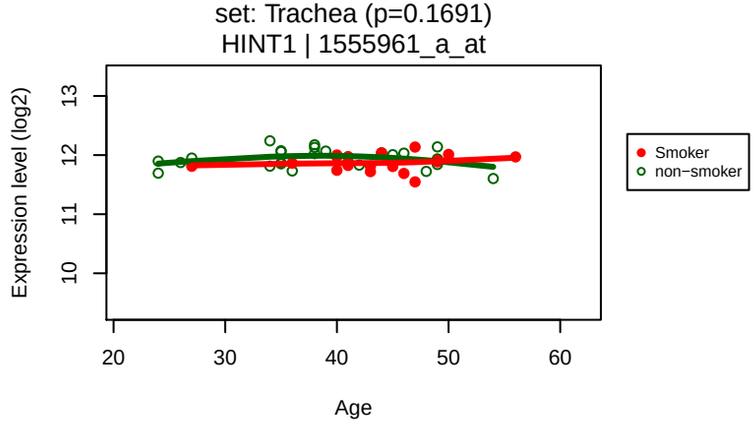

Gene expression ratio in Males
gene: HINT1 | 1555961_a_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6156
p=0.3085
p=0.0167
p=0.6407
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: HINT1 | 1555961_a_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1028
p=0.4168
p=0.9299
p=0.0836
SmallAirway2
LargeAirway2
Trachea
Macrophage2

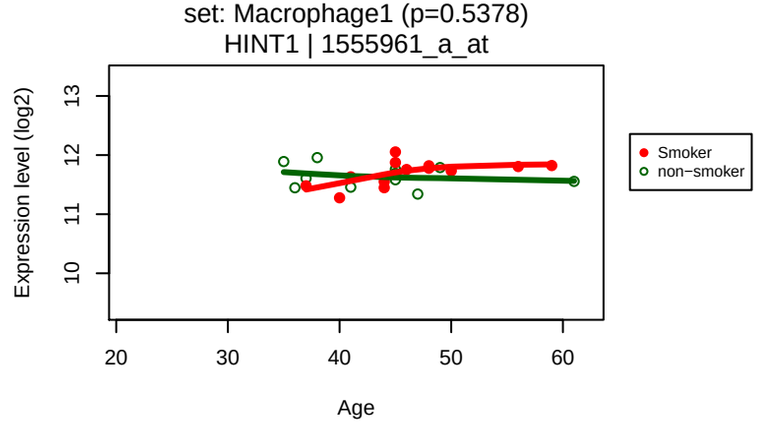

set: SmallAirway1 (p=0.0022)
HINT1 | 1555961_a_at
Expression level (log2)
10
11
12
13
20
30
40
50
60
Age
Smoker
non−smoker

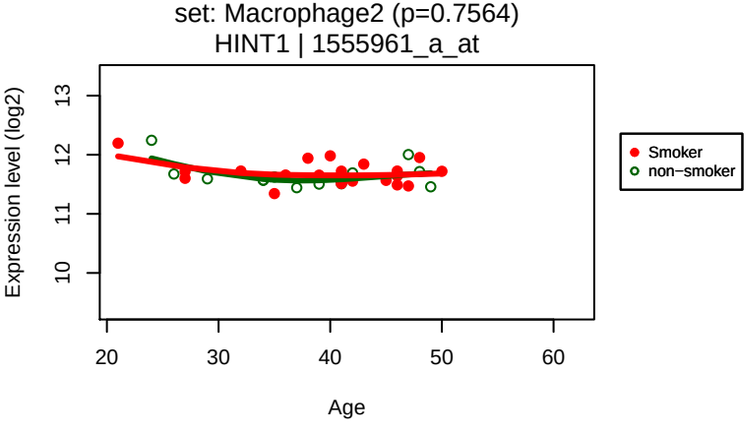

set: SmallAirway2 (p=0.9502)
HINT1 | 1555961_a_at
Expression level (log2)
10
11
12
13
20
30
40
50
60
Age
Smoker
non−smoker

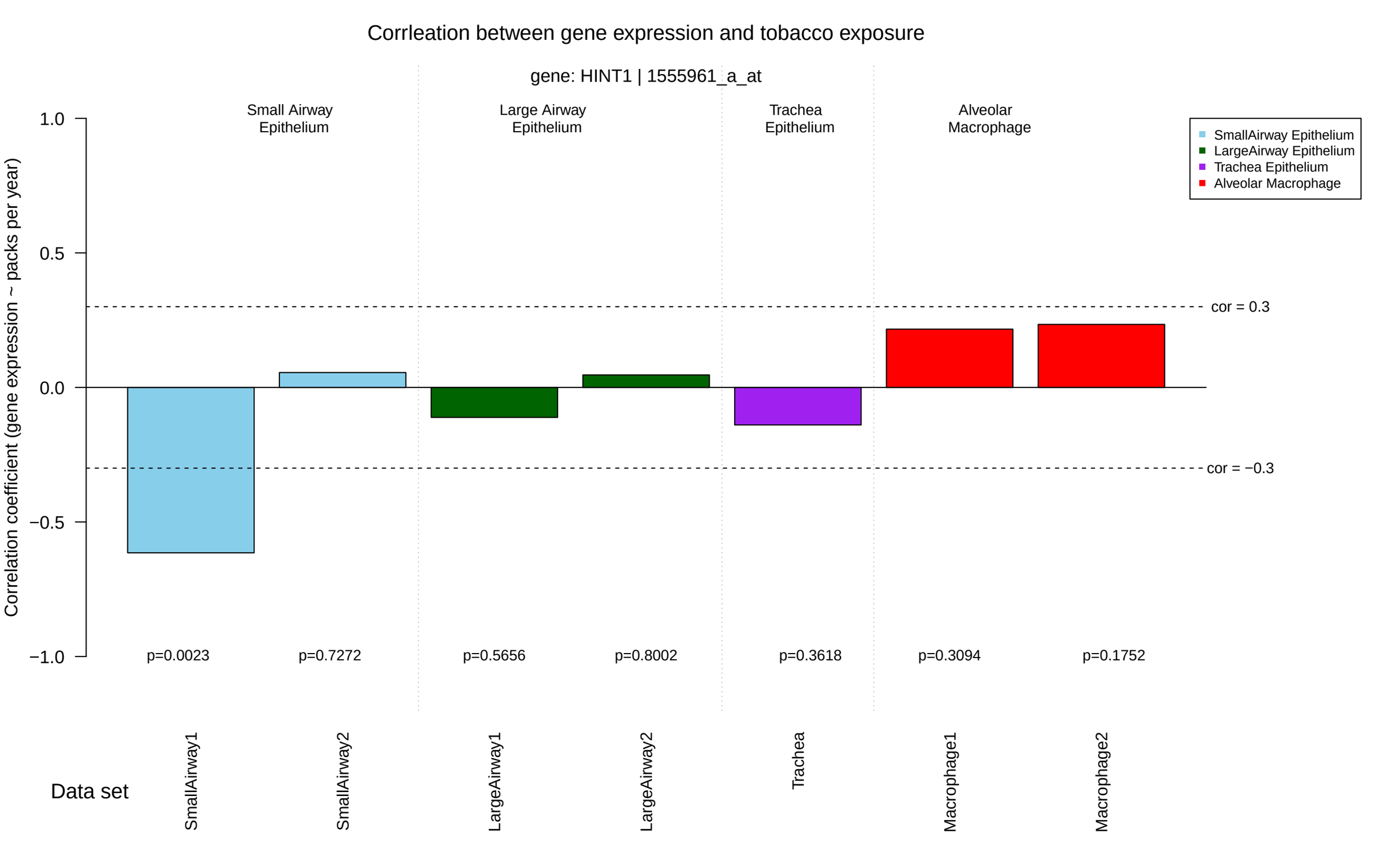

set: LargeAirway1 (p=0.0108)
HINT1 | 1555961_a_at
Expression level (log2)
10
11
12
13
20
30
40
50
60
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.5317)
HINT1 | 1555961_a_at
Expression level (log2)
10
11
12
13
20
30
40
50
60
Age
Smoker
non−smoker


set: Trachea (p=0.1691)
HINT1 | 1555961_a_at
Expression level (log2)
10
11
12
13
20
30
40
50
60
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5378)
HINT1 | 1555961_a_at
Expression level (log2)
10
11
12
13
20
30
40
50
60
Age
Smoker
non−smoker


set: Macrophage2 (p=0.7564)
HINT1 | 1555961_a_at
Expression level (log2)
10
11
12
13
20
30
40
50
60
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: HINT1 | 1555961_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0023
p=0.7272
p=0.5656
p=0.8002
p=0.3618
p=0.3094
p=0.1752
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2